TEAMECHO · GALAXY AI SPARTANS HACKATHON · IMMIGRATION TRACK

Immigration Case Co-Pilot

An agentic case operations co-pilot for an immigration law firm. It turns messy intake, client messages and uploaded PDFs into a traceable, review-ready case. Software prepares, legal staff approve, and nothing is ever filed or sent.

The problem today

- Case files arrive as a pile of PDFs, scans, Zoom notes and chat messages with no structure.
- Paralegals sort, rename and check every file by hand, and duplicates or another client's papers slip in.
- Details disagree across sources, and a value from a chat message can quietly become a "fact".
- Deadlines are copied from calendars instead of the government notice that actually sets them.
- There is no reliable record of who decided what, when and on which evidence.

What the Co-Pilot does

- Reads every document from its content, cites a quote and page for each value, and checks the quote by code.
- Sets aside duplicates, old versions and wrong-client files on its own, each with a written reason.
- Keeps every contradiction visible with both sources cited and opens a review for a person.
- Computes deadlines by code from the verified portal event only, using the firm's rule.
- Routes every change through propose, legal review and commit, recorded on a tamper-evident ledger.

Faster intake

Staff start from a short exception list instead of a folder of files. Classification, duplicate checks and checklists run on their own, and each case shows why it is blocked.

Fewer costly mistakes

Claims never overwrite evidence. Wrong-client papers are quarantined. Unknown values stay blank. Deadlines come only from verified events, so a typed calendar date cannot set one.

Defensible record

Every approval, rejection and denied attempt is written to a hash-chained ledger. One click proves the history was not edited, which supports supervision and compliance reviews.

All data is synthetic. Labels such as OPS-01 and LEGAL-01 are demo controls, not authentication.

BUSINESS VALUE

Who gains what

The Co-Pilot does the preparation work and leaves every legal judgment and every government-facing action with people.

Area	Today, by hand	With the Co-Pilot
Document intake	Open each file, guess its type from the name, rename and sort it.	Type read from content. Duplicates, superseded copies and wrong-client files set aside with a reason. Uploaded is kept separate from usable.
Fact checking	Compare passport, letters, notes and messages line by line.	One fact table with source, quote and page per value. Disagreements become cited conflicts for review.
Missing evidence	Discovered late, often when a form is being prepared.	Checklist per case from the document rules. Missing items show as the queue reason from day one.
Deadlines	Copied from calendars or emails, easy to mistype.	Computed by code from the verified portal event plus the firm's rule, then approved by legal before it is active.
Client replies	Written from scratch, risk of promising an outcome.	Grounded draft that refuses to predict outcomes, marked DRAFT, NOT SENT.
Supervision	Hard to show who approved what and on which evidence.	OPS proposes, LEGAL approves or rejects, OPS commits. Every step and every denied attempt is on the ledger.
AI governance	Prompts hidden in code, changed by developers only.	Prompts, models and schemas are versioned in the app, tested on sample text and rolled back in one click.
AI cost	One large model for every task, whatever the cost.	Start on Haiku, the fastest and lowest-cost model, and switch a prompt to Sonnet only when accuracy needs it. No code change.

Model choice: start small, step up only when needed

Claude Haiku
Starting point. Sorts every upload by type. Fast and lowest cost, so it handles the high-volume step.

switch if accuracy drops →

Claude Sonnet
Reads document fields with a quote and page. Any Haiku prompt can be moved here in the AI management screen.

only for analysis →

Claude Opus
Kept for the few steps that need deep reasoning, such as explaining a case's conflicts.

The model is a setting on each prompt in the registry, not in code. Every run records the model and token counts, so the cost of each switch is visible.

Operations staff (OPS)

Less sorting and chasing. A queue that explains itself. One place to propose stage changes, forms, packets and replies.

Attorneys (LEGAL)

Review only what needs a decision, with both sides of every conflict cited. Approve or reject with a reason that a rerun can never undo.

Firm management

AI that is grounded, logged and governed. No live integrations and nothing sent outside the firm, which keeps the risk low while adoption grows.

Immigration Case Co-Pilot

Messy intake in, review-ready case out. Claude reads, code decides, people approve. Every step is audited.

Claude (words and JSON) · Deterministic code · Human decision · Storage · Service or input

Firm staff
OPS-xx prepares and proposes
LEGAL-xx approves or rejects
"Acting as" is a demo control

React 19 frontend — Vite, TypeScript, Tailwind 4, TanStack Query. Every value shows a source chip and a status pill with a reason.

Upload · Case queue · Documents · Facts, conflicts · Review queue · Ask the case · AI management · History, /status

/api with X-Actor header, server enforces permissions

Inputs (synthetic)

- **PDF uploads** — EVIDENCE — passports, pay slips, letters
- **Portal events** — EVIDENCE — verified MOCK_PORTAL only
- **Client profiles** — EVIDENCE — human-verified records
- **Messages** — CLAIM — email, WhatsApp exports
- **Zoom notes, calendar** — CLAIM — what someone said

All content is untrusted data. Instructions inside are flagged, never followed.

FastAPI backend

Python 3.12, SQLAlchemy 2, Pydantic 2. Three trust zones. Long work runs as background jobs.

1. Claude reads and writes words

- **Prompt registry** — versioned prompts in the database, no prompts in code
- **Document reader** — classify, then read_passport, read_pay_statement ...
- **Conflict explainer** — plain-word explanation that cites both sides
- **Client triage** — labels incoming messages, replies come from templates
- **Case investigation agent** — read-only tools, case_id fixed, answers with citations

Structured JSON only. Every value carries a quote and page. No write tools. Runs logged with prompt version and tokens.

JSON →

← explain

2. Code decides facts and rules

- **Ingest and scope** — sha256 duplicates, page owner check, quarantine
- **Quote check** — value is verified only if its quote is on the page
- **Rules and deadlines** — document rules, cross checks, portal event + 10 days
- **Fact engine** — one fact per source, conflicts per field, claim vs evidence
- **Proposal builder** — checklist, form draft, deadline, reply, idempotency key

Permissions, case scoping and date math live here. Unknown stays unknown. Nothing is invented.

proposals →

3. People make every legal call

- **OPS proposes** — status PENDING_LEGAL_REVIEW
- **LEGAL reviews** — sees quotes, sources and conflicts side by side
  - APPROVED
  - REJECTED
  - SENT_BACK
- **OPS commits** — refused unless APPROVED, then COMMITTED

A rejection is permanent until a person changes it. Reruns never overwrite it. Nothing is filed or sent to anyone.

- **Starter kit** — synthetic cases, profiles, messages, rules — *read-only mount*
- **Anthropic API** — Sonnet 5.5 reads documents; Opus 5.5 analyses the case — *cached by sha256 + prompt version*
- **PostgreSQL + pgvector** — cases, documents, facts, proposals, prompts, runs — *filtered by case_id first*
- **Audit ledger** — hash-chained, insert only, one chain per case — *any edit breaks verify*

One case, end to end

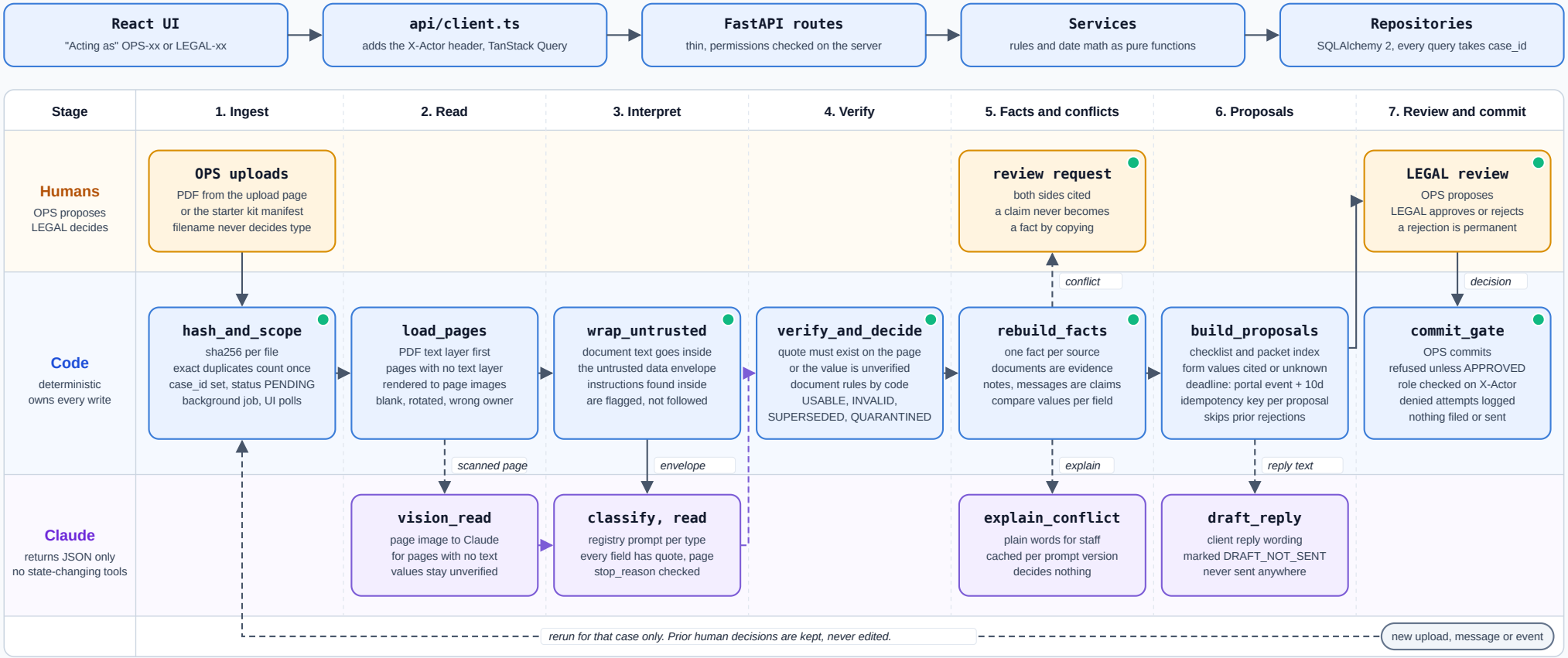
A new upload, message or event reruns the case. Human decisions stay.

Ingest → Read pages → Claude reads → Verify quotes → Facts → Conflicts → Proposals → LEGAL review → Commit

- **No legal advice** — no eligibility or outcome predictions
- **Nothing leaves the building** — DRAFT_NOT_SENT, NOT FOR FILING
- **Claims never become facts** — conflicts go to a person with both sources
- **Never crosses cases** — every query, search and chat takes case_id
- **Every change is audited** — denied attempts too, in the same transaction

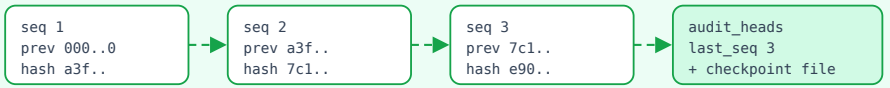
Immigration Case Co-Pilot: workflow architecture

How a case moves from intake to a committed, audited decision. Claude reads and drafts, code decides facts and permissions, humans decide anything legal.



Audit ledger (Postgres, insert only, hash-chained)

Every green dot writes one line through audit.record(), in the same transaction as the change.
One chain per case plus a SYSTEM chain. A trigger blocks UPDATE, DELETE and TRUNCATE.
GET /api/cases/{id}/audit/verify recomputes the chain and reports the first broken seq.



Shared services and stores

PostgreSQL + pgvector
uploads, facts, conflicts, proposals
search filtered by case_id first

Prompt registry
seeded from ai/prompts/*.yaml, versioned
model and effort live here, not in code

prompt_runs
key, version, model, tokens, input hash
cache by file sha256 + prompt version

Local embeddings
semantic search over case text
downloaded at build, works offline

Investigate agent
"Why is this blocked?" panel
read-only tools, bounded rounds

Legend

Human decision · Deterministic code · Claude call, JSON only · Writes an audit line
main flow · conditional · Claude output back to code

Guarantees

No legal advice, no filing, no sending. Drafts are DRAFT_NOT_SENT, forms NOT FOR FILING.
Every query is scoped by case_id. Unclear ownership means the document is quarantined.
A value is verified only when code finds its quote in the source text.
Deadlines come only from a verified portal event plus 10 days, computed by code.

Edge case handling

Where each trap is caught in the pipeline, and what the system does instead of guessing. Case ids come from the synthetic starter kit.

Claude reads and drafts · Deterministic code · People decide · Refused, held or flagged · Sent to a person · Handled and kept · Set aside or left blank · RULE Applies to every case

1 Upload and scope

Code

CASE-0001 DUPLICATE
Same file uploaded twice
Exact sha256 match counts once.
Second copy set aside with a reason.

RULE REFUSED
Zip, empty or fake PDF
Screened before upload.
Each file left out with a reason.

CASE-0003 QUARANTINED
Wrong client's document
Owner read from content, not the folder name. No data crosses cases.

CASE-0026 QUARANTINED
Mixed PDF, two clients
Only the foreign page is held.
The rest of the file is kept.

CASE-0005 SUPERSEDED
Older version of a document
Old copy marked superseded.
The newest version is accepted.

2 Read the document

Claude reads, code checks

CASE-0004 NEEDS REVIEW
Unreadable passport
Marked not usable. A person reviews it. Nothing is guessed.

CASE-0025 NEEDS REVIEW
Rotated scan
OCR fallback first, then sent to human review when unclear.

CASE-0008 TRANSLATION
Letter in another language
Marked as needing translation.
Not used as evidence yet.

CASE-0021 NEEDS REVIEW
Translation with no original
Orphan translation is held for review and not counted.

CASE-0017 FLAGGED
Hidden instruction in a PDF
Wrapped as untrusted data.
Flagged and logged, never followed.

RULE UNVERIFIED
Quote not in the source
Code searches the source text.
No match means unverified.

RULE NOT READ YET
AI service down or refusal
The file stays safe and is marked for review. Retry later.

3 Facts and conflicts

Code

CASE-0001 CONFLICT
Name: Dawan vs Dhawan
Both sides cited with quote and page. Review opens, no silent pick.

CASE-0002 CONFLICT
Date of birth off by one day
1991-03-05 vs 1991-03-04 is kept as a conflict for LEGAL.

CASE-0013 NEEDS PROOF
Address change in a message
A message is a claim. The field is never updated without proof.

CASE-0023 CONFLICT
Start date: Zoom vs profile
Zoom note says 2025, profile 2023.
Claim and evidence both cited.

CASE-0016 UNVERIFIED
Calendar says APPROVED
Status stays Processing.
A calendar entry is not evidence.

RULE BLANK
No evidence for a form field
Unknown stays blank with a reason.
No value is invented.

4 Checklist and deadline

Code

CASE-0014 MISSING
No passport uploaded
Checklist item shows Missing.
Reason shown on the case queue.

CASE-0006 MISSING
September payslip missing
The checklist names the exact item that is still missing.

CASE-0024 MISSING
Prior notice missing
Required item flagged as not yet satisfied.

CASE-0007 EXPIRED
Passport expired 2026-09-30
Expiry checked by code.
Not counted as valid identity.

CASE-0009 CONFLICT
Calendar vs portal deadline
Verified portal event + 10 days.
Calendar date flagged for LEGAL.

CASE-0015 OVERDUE
Deadline already passed
2026-10-15 shown as overdue with the rule that set it.

RULE IGNORED
Calendar entry as deadline
Never creates a deadline.
Shown as Calendar, not evidence.

5 Drafts and chat

Claude drafts, code limits

CASE-0010 NO PREDICTION
Will I be approved?
The reply refuses to predict.
Marked DRAFT, NOT SENT.

CASE-0019 REFUSED
Chat about another client
Retrieval filtered by case_id.
Question refused and logged.

RULE NOT FOR FILING
Form draft generated
Every field cited to a source.
PDF stamped NOT FOR FILING.

CASE-0020 FLAGGED
Payment amount mismatch
The mismatch is shown for a person to resolve.

6 Review and commit

People decide, code enforces

RULE DENIED
OPS tries to approve
The server answers 403.
The denied attempt is logged.

RULE DENIED
Commit before approval
Refused unless the proposal state is APPROVED.

RULE REQUIRED
Reject with no reason
LEGAL must give a reason.
A blank reason is refused.

RULE KEPT
Rerun after a rejection
The rejection stays. The skipped rerun is logged on the ledger.

RULE SAME RESULT
Same idempotency key twice
Returns the same proposal.
A key from another case is refused.

CASE-0012 KEPT
Re-import after a human fix
The correction Meera Batra stays.
Same snapshot stored once.

RULE 401
Missing or unknown actor
Refused before any case data is read.

Audit ledger, under every column

Every outcome above, including DENIED, QUARANTINED and FLAGGED, writes one hash-chained audit line in the same transaction as the change.

Verify chain recomputes every hash and reports the first broken seq. Entries hold ids, hashes and states, never document text.

EDGE CASE HANDLING

Control, permission and AI safety guards

These apply to every case. The server enforces them even when the screen would allow the click.

Situation	What the system does	Outcome
OPS tries to approve or reject	The server answers 403. The denied attempt is written to the ledger.	DENIED
Commit before approval	Commit is refused unless the proposal state is APPROVED.	DENIED
Approved stage change gone stale	Commit is refused with STAGE_STALE when the case has moved on since the proposal. Found and fixed during the 100-case run.	DENIED
Rejection with no reason	Legal must give a reason. A blank reason is refused.	REQUIRED
Rerun after a legal rejection	The rejection stays until a person changes it. The skipped rerun is logged. Verified on CASE-0070.	KEPT
Same request sent twice	The same idempotency key returns the same proposal. A key from another case is refused.	SAME RESULT
Missing or unknown actor	Refused with 401 before any case data is read.	401
User not assigned to the case	Refused with CASE_NOT_ASSIGNED. Every query takes a case_id.	403
Bad upload: zip, empty, oversized, fake PDF	Screened before upload. Each file is left out with a reason.	REFUSED
Similar client name on a new case	A possible match warning lists the existing cases before a new one is created.	POSSIBLE MATCH
AI quote not found in the source	Code searches the extracted text. No match means the value is unverified.	UNVERIFIED
AI service down, refusal or failure	stop_reason is checked first. The file stays safe, is marked for human review and can be read again later.	NOT READ YET
Someone edits the audit history	Each line is linked to the previous one by SHA-256. Verify chain reports the first broken seq. A trigger blocks UPDATE, DELETE and TRUNCATE.	BREAK DETECTED